

Hatcher Chapter 3: Cohomology

Arham Lodha

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1 Universal Coefficient Theorem for Cohomology

2 Cup Product

2.1 Problems

Problem 3.2.5. Show the ring $H^*(\mathbb{R}P^\infty, \mathbb{Z}_{2k}) \cong \mathbb{Z}_{2k}[\alpha, \beta]/(2\alpha, 2\beta, \alpha^2 - k\beta)$

Proof. By Theorem 3.19, $H^*(\mathbb{R}P^\infty, \mathbb{Z}_2) = \mathbb{Z}[a]$. Use standard cellular structure and standard cellular cohomology to see that

$$H^0(\mathbb{R}P^n, \mathbb{Z}_{2k}) = \mathbb{Z}_{2k}$$

,

$$H^n(\mathbb{R}P^n, \mathbb{Z}_{2k}) = \mathbb{Z}_2, \forall n \geq 1$$

Let $r : \mathbb{Z}_{2k} \rightarrow \mathbb{Z}_2$ be the Coefficient maps reduction mod 2 ie $[x]_{2k} \rightarrow [x]_2$. It induces a surjective ring map, $r^* : H^*(\mathbb{R}P^\infty, \mathbb{Z}_{2k}) \rightarrow H^*(\mathbb{R}P^\infty, \mathbb{Z}_2)$

Let $\alpha \in H^1(\mathbb{R}P^\infty, \mathbb{Z}_{2k})$ where $r^*(\alpha) = a \in H^1(\mathbb{R}P^\infty, \mathbb{Z}_2) = \mathbb{Z}[a]$, since α maps to the generator (only non 0 element) it must also be nonzero. Since $H^1(\mathbb{R}P^\infty, \mathbb{Z}_{2k}) = \mathbb{Z}_2$, α generates H^1 .

Let $\beta \in H^2(\mathbb{R}P^\infty, \mathbb{Z}_{2k})$ where $r^*(\beta) = a^2$ and β generates $H^2(\mathbb{R}P^\infty, \mathbb{Z}_{2k})$.

Note since $H^n = \mathbb{Z}_2$, $2\alpha = 0$ and $2\beta = 0$. So the $(2\alpha, 2\beta)$ is forced on the additive structure.

Now lets see how α^2 and β relate.

Take the short exact sequence

$$0 \rightarrow \mathbb{Z}_k \xrightarrow{i} \mathbb{Z}_{2k} \rightarrow \mathbb{Z}_2 \rightarrow 0$$

The Bockstein Homomorphisms induced by the coefficient sequence:

$B(a) = a^2 = k\beta$. So naturality gives $r^*(\alpha) = a^2 = r^*(k\beta) \implies \alpha - k\beta = 0$. So the ideal $(\alpha - k\beta)$ gets added.

Thus $H^*(\mathbb{R}P^\infty, \mathbb{Z}_{2k}) = \mathbb{Z}_{2k}[\alpha, \beta]/(2\alpha, 2\beta, \alpha^2 - k\beta)$, where $|\alpha| = 1$ and $|\beta| = 2$. $\square$

Problem 3.2.6. Use cup products to compute the map $H^*(\mathbb{C}P^n, \mathbb{Z}) \rightarrow H^*(\mathbb{C}P^n, \mathbb{Z})$ induced by the map $\mathbb{C}P^n \rightarrow \mathbb{C}P^n$ that is the quotient of the map $\mathbb{C}^{n+1} \rightarrow \mathbb{C}^{n+1}$ raising each coordinate to the d th power.

Proof. Let $P^n = \mathbb{C}P^n$ and all coefficients are $\mathbb{Z}$ unless otherwise stated.

1. For $n = 1$: $\mathbb{C}P^1 = S^2$, by Riemann Sphere. Thus the map $H^*(S^2) \rightarrow H^*(S^2)$ is just the $\times d$ map on H^2 and 0 map everywhere else.
2. For $n > 1$: $i : \mathbb{C}P^1 \rightarrow \mathbb{C}P^n$ be the natural inclusion.

$$\begin{array}{ccc} H^2(P^n) & \xrightarrow{d} & H^2(P^n) \\ i^* \downarrow & & \downarrow i^* \\ H^1(P^1) & \xrightarrow{d} & H^1(P^1) \end{array}$$

i^* is a isomorphism. So top d must be $d \times$.

□

Problem 3.2.7. Use cup products to show $\mathbb{R}P^3$ is not homotopy equivalent to $\mathbb{R}P^2 \vee S^3$

Proof. 1. **Without cup products:** Universal cover of P^3 is S^3 and universal cover of $P^2 \vee S^3 = S^3 \vee S^2 \vee S^3$. $H_3(S^3) = \mathbb{Z}$ and $H_3(S^3 \vee S^2 \vee S^3) = \mathbb{Z} \oplus \mathbb{Z}$. Thus universal covers are not homeomorphic. Thus the spaces are not homotopy equivalent.

2. **With Cup products:** Unless explicitly stated $\mathbb{Z}_2$ coefficients are being used. By Theorem 3.19:

$$H^*(P^3) := \mathbb{Z}_2[\alpha]/(\alpha^4)$$

By Cellular Cohomology and 3.14:

$$H^*(P^2 \vee S^3) = \frac{\mathbb{Z}_2[\alpha]}{(\alpha^3)} \times \frac{\mathbb{Z}_2[\beta]}{(\beta^2)}$$

where $|\alpha| = 1$ and $|\beta| = 3$. Every degree 1 class has the form, $(\lambda\alpha, 0) \implies (\lambda\alpha, 0)^3 = (\lambda^3\alpha^3, 0) = (0, 0) = 0$. Thus no element of degree 1 has a nonzero cube. Thus both graded rings are not isomorphic. Thus both spaces are not homotopy equivalent.

□

Problem 3.2.8. Let $X = \mathbb{C}P^2$ with a cell e^3 attached by a map $S^2 = \mathbb{C}P^1 \subset \mathbb{C}P^2$ with degree p , and let $Y = M(\mathbb{Z}_p, 2) \vee S^4$. Thus X and Y have the same 3 skeleton but differ in the way their 4 cells are attached. Show that X and Y have isomorphic cohomology rings with $\mathbb{Z}$ coefficients but not with $\mathbb{Z}_p$ coefficients.

Proof. 1. $\mathbb{Z}$ coefficients:

- (a) Cohomology of X and Y :

$$0 \longrightarrow \mathbb{Z} \longrightarrow 0 \longrightarrow \mathbb{Z} \xrightarrow{\times p} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \longrightarrow 0$$

$$H^n(X) := \begin{cases} \mathbb{Z} & \text{if } n = 0, 4 \\ \mathbb{Z}_p & \text{if } n = 3 \\ 0 & \text{if } n = 1, 2 \text{ or } n > 4 \end{cases}$$

It makes sense that for degree reasons that the cup product structure is trivial. (Cup product of any two nonzero homology generators)

2. $\mathbb{Z}_p$ coefficients

(a) Cohomology of X :

$$0 \longrightarrow \mathbb{Z}_p \longrightarrow 0 \longrightarrow \mathbb{Z}_p \xrightarrow{\times p} \mathbb{Z}_p \xrightarrow{0} \mathbb{Z}_p \longrightarrow 0$$

$$H^n(X) := \begin{cases} \mathbb{Z}_p & \text{if } n = 0, 2, 3, 4 \\ 0 & \text{if } n = 1 \text{ or } n > 4 \end{cases}$$

The cup product is only nontrivial on the 2 chain. So the cup product structure reduces to the cup product structure of $\mathbb{C}P^2$. Let $\alpha \in H^2(X)$, be the generator by Theorem 3.19, $\alpha \smile \alpha$ is the generator of $H^4(X)$.

(b) Cohomology of Y

$$0 \longrightarrow \mathbb{Z}_p \longrightarrow 0 \longrightarrow \mathbb{Z}_p \xrightarrow{\times p} \mathbb{Z}_p \xrightarrow{0} \mathbb{Z}_p \longrightarrow 0$$

$$H^n(X) := \begin{cases} \mathbb{Z}_p & \text{if } n = 0, 2, 3, 4 \\ 0 & \text{if } n = 1 \text{ or } n > 4 \end{cases}$$

The reduced Cohomology ring is:

$$\tilde{H}^*(Y) := \tilde{H}^*(M(\mathbb{Z}_p, 2)) \times \mathbb{Z}[\beta]/[\beta^2]$$

for $|\beta| = 4$. Here the generator of the H^2 , cup product with itself will be trivial.

□

Problem 3.2.9. Show that if $H_n(X, \mathbb{Z})$ are free for all n , then $H^*(X, \mathbb{Z}_p)$ and $H^*(X, \mathbb{Z}) \otimes \mathbb{Z}_p$ are isomorphic as rings, so in particular the ring structure with $\mathbb{Z}$ coefficients determines the ring structure with $\mathbb{Z}_p$ coefficients.

Proof. By Universal Coefficient Theorem for Homology (Theorem 3A.3),

$$H_n(X, \mathbb{Z}_p) = H_n(X, \mathbb{Z}) \otimes \mathbb{Z}_p \oplus \text{Tor}(H_{n-1}(X, \mathbb{Z}), \mathbb{Z}_p) = H_n(X, \mathbb{Z}) \otimes \mathbb{Z}_p$$

The last equality is true because $H_n(X)$ is free for all n , furthermore $H_n(X) \otimes \mathbb{Z}_p = \mathbb{Z}_p^{\text{rank } H_n(X)}$ ie a free module over $\mathbb{Z}_p$.

By Universal Coefficient Theorem for Cohomology (Theorem 3.2):

$$H^n(X, \mathbb{Z}_p) = \text{Ext}(H_{n-1}(X, \mathbb{Z}_p), \mathbb{Z}_p) \oplus \text{Hom}(H_n(X, \mathbb{Z}_p), \mathbb{Z}_p) = \text{Hom}(H_n(X, \mathbb{Z}_p), \mathbb{Z}_p) = \mathbb{Z}_p^{\text{rank } H_n(X)}$$

Since $H_{n-1}(X, \mathbb{Z}_p)$ is free over $\mathbb{Z}_p$, Ext term disappears.

$$H^n(X, \mathbb{Z}) = \text{Ext}(H_{n-1}(X, \mathbb{Z}), \mathbb{Z}) \oplus \text{Hom}(H_n(X, \mathbb{Z}), \mathbb{Z}) = \mathbb{Z}^{\text{rank} H_n(X)}$$

Thus $H^n(X, \mathbb{Z}) \otimes \mathbb{Z}_p = \mathbb{Z}_p^{\text{rank} H_n(X)} \cong H^n(X, \mathbb{Z}_p)$. Thus there is a graded ring isomorphism.

Let $\pi : \mathbb{Z} \rightarrow \mathbb{Z}_p$ standard mod map. Let $\rho_p : C^*(X, \mathbb{Z}) \rightarrow C^*(X, \mathbb{Z}_p)$ be the standard coefficient reduction map where $c \rightarrow \pi \circ c$. π is a ring map. Let $\alpha \in C^p(X, \mathbb{Z})$, $\beta \in C^q(X, \mathbb{Z})$ and σ be a $(p+q+1)$ simplex,

$$\begin{aligned} \rho_p(\alpha \smile \beta)(\sigma) &= (\pi \circ (\alpha \smile \beta))(\sigma) \\ &= \pi(\alpha(\sigma|_{[v_0, \dots, v_p]})\beta(\sigma|_{[v_{p+1}, \dots, v_{p+q}]})) \\ &= \pi(\alpha(\sigma|_{[v_0, \dots, v_p]}))\pi(\beta(\sigma|_{[v_{p+1}, \dots, v_{p+q}]})) \\ &= (\pi \circ \alpha)(\sigma|_{[v_0, \dots, v_p]})(\pi \circ \beta)(\sigma|_{[v_{p+1}, \dots, v_{p+q}]}) \\ &= \rho_p(\alpha)(\sigma|_{[v_0, \dots, v_p]})\rho_p(\beta)(\sigma|_{[v_{p+1}, \dots, v_{p+q}]}) \\ &= (\rho_p(\alpha) \smile \rho_p(\beta))(\sigma) \end{aligned}$$

Thus $\rho_p(\alpha \smile \beta) = \rho_p(\alpha) \smile \rho_p(\beta)$.

$$\rho_p^* : H^*(X, \mathbb{Z}) \rightarrow H^*(X, \mathbb{Z}_p)$$

is a ring homomorphism. Let

$$\varphi_p = \rho_p^* \otimes 1_{\mathbb{Z}_p} : H^*(X, \mathbb{Z}) \otimes_{\mathbb{Z}} \mathbb{Z}_p \rightarrow H^*(X, \mathbb{Z}_p)$$

φ_p is the graded abelian group iso seen above. Tensoring with $\mathbb{Z}_p$ doesn't change the ring homomorphism, thus φ_p is a ring isomorphism. $\square$

Problem 3.2.10. Show that the cross product map $H^*(X, \mathbb{Z}) \otimes H^*(Y, \mathbb{Z}) \rightarrow H^*(X \times Y, \mathbb{Z})$ is not a isomorphism if X and Y are infinite discrete sets.

Proof. Note $H^*(X, \mathbb{Z}) = \Pi_{x \in X} \mathbb{Z}$ similarly for Y . $H^*(X \times Y, \mathbb{Z}) = \Pi_{(x,y) \in X \times Y} \mathbb{Z}$.

$$\Pi_{x \in X} \mathbb{Z} \otimes \Pi_{y \in Y} \mathbb{Z} \subset \Pi_{(x,y) \in X \times Y} \mathbb{Z}$$

where the inclusion is the natural one,

$$(n_x)_{x \in X} \otimes (m_y)_{y \in Y} \rightarrow (n_x m_y)_{x \in X, y \in Y}$$

,

UNFINISHED

$\square$

Problem 3.2.11. Using cup products show that every map $S^{m+n} \rightarrow S^m \times S^n$ induces the trivial homomorphism $H_{m+n}(S^{m+n}) \rightarrow H_{m+n}(S^m \times S^n)$, assuming $m > 0$ and $n > 0$.

Proof. It is known

$$H^k(S^n) := \begin{cases} \mathbb{Z} & \text{if } k = 0, n \\ 0 & \text{otherwise} \end{cases}$$

$\forall f : S^{m+n} \rightarrow S^m \times S^n$. By Proposition 3.10, f induces

$$f^* : H^*(S^m \times S^n) \rightarrow H^*(S^{m+n})$$

By Theorem 3.15, $H^i(S^n) \otimes_{\mathbb{Z}} H^j(S^m) \cong H^{i+j}(S^n \times S^m)$ and $p_1(a) \smile p_2(b)$ is a generator for $H^{i+j}(S^n \times S^m)$ where a and b are generators. $\forall (a, b) \in H^i(S^n) \otimes_{\mathbb{Z}} H^j(S^m)$.

$$\begin{aligned} f^{i+j}(p_1^i(a) \smile p_2^j(b)) &= f^i(p_1^i(a)) \smile f^j(p_2^j(b)) \\ &= (p_1 \circ f)^i(a) \smile (p_2 \circ f)^j(b) \end{aligned}$$

we get the following commutative diagram for all $0 < i, j < n$

$$\begin{array}{ccccc} H^i(S^{m+n}) & \times & H^j(S^{m+n}) & \xrightarrow{\smile} & H^{i+j}(S^{m+n}) \\ \uparrow (p_1 \circ f)^i & & \uparrow (p_2 \circ f)^j & & \uparrow f^{i+j} \\ H^i(S^m) & \otimes_{\mathbb{Z}} & H^j(S^n) & \xrightarrow{\times} & H^{i+j}(S^m \times S^n) \end{array}$$

Note $\times$ is an isomorphism. Furthermore note that $(p_1 \circ f)^i$ and $(p_2 \circ f)^j$ are trivial because $H^i(S^{m+n})$ and $H^j(S^{m+n})$ are 0. Thus f^{i+j} must be 0. Thus $f^{m+n} = 0$.

By the Universal Coefficient Theorem for Cohomology we get the following:

$$\begin{array}{ccccccc} 0 & \longrightarrow & \text{Ext}(H_{m+n-1}(S^m \times S^n), \mathbb{Z}) & \longrightarrow & H^{m+n}(S^m \times S^n) & \longrightarrow & \text{Hom}(H_{m+n}(S^m \times S^n), \mathbb{Z}) \longrightarrow 0 \\ & & \downarrow (f_*)^* & & \downarrow f^*=0 & & \downarrow (f_*)^* \\ 0 & \longrightarrow & \text{Ext}(H_{m+n-1}(S^{m+n}), \mathbb{Z}) & \longrightarrow & H^{m+n}(S^{m+n}) & \longrightarrow & \text{Hom}(H_{m+n}(S^{m+n}), \mathbb{Z}) \longrightarrow 0 \end{array}$$

all the Exts are 0, because all homology groups are free. Thus 3 arrows in each short exact sequence is an isomorphism. Thus if $f^* = 0 \implies (f_*)^* = 0 \implies f_* = 0$. $\square$

Problem 3.2.12. Show that the spaces $(S^1 \times \mathbb{C}P^\infty)/(S^1 \times \{x_0\})$ and $S^3 \times \mathbb{C}P^\infty$ have isomorphic cohomology rings with $\mathbb{Z}$ and any other coefficients.

Proof. $\mathbb{Z}$ **coefficients:** By Example 3.14 and Theorem 3.19,

$$H^*(S^3 \times \mathbb{C}P^\infty) \cong \frac{\mathbb{Z}[\alpha]}{\alpha^2} \otimes_{\mathbb{Z}} \mathbb{Z}[\beta] \cong \frac{\mathbb{Z}[\alpha, \beta]}{\alpha^2}$$

where $|\alpha| = 3$ and $|\beta| = 2$.

By Excision ,

$$\tilde{H}^k((S^1 \times \mathbb{C}P^\infty)/(S^1 \times \{x_0\})) = H^k(S^1 \times \mathbb{C}P^\infty, S^1 \times \{x_0\}) = H^k(S^1 \times \mathbb{C}P^\infty, \emptyset \times \mathbb{C}P^\infty \cup S^1 \times \{x_0\})$$

By Relative Kunneth's Formula (Theorem 3.18),

$$\begin{aligned}
H^k(S^1 \times \mathbb{C}P^\infty, \emptyset \times \mathbb{C}P^\infty \cup S^1 \times \{x_0\}) &= \bigoplus_{n=0}^k H^n(S^1, \emptyset) \otimes_{\mathbb{Z}} H^{k-n}(\mathbb{C}P^\infty, \{x_0\}) \\
&= \bigoplus_{n=0}^k H^n(S^1) \otimes_{\mathbb{Z}} \tilde{H}^{k-n}(\mathbb{C}P^\infty) \\
&= \mathbb{Z} \otimes_{\mathbb{Z}} \tilde{H}^k(\mathbb{C}P^\infty) \oplus \mathbb{Z} \otimes_{\mathbb{Z}} \tilde{H}^{k-1}(\mathbb{C}P^\infty)
\end{aligned}$$

Thus

$$H^k(S^1 \times \mathbb{C}P^\infty, \emptyset \times \mathbb{C}P^\infty \cup S^1 \times \{x_0\}) = \begin{cases} 0 & \text{if } k \leq 1 \\ \mathbb{Z} & \text{otherwise} \end{cases}$$

Thus

$$H^k((S^1 \times \mathbb{C}P^\infty)/(S^1 \times \{x_0\})) = \begin{cases} 0 & \text{if } k = 1 \\ \mathbb{Z} & \text{otherwise} \end{cases}$$

Let $a \in H^1(S^1)$ and $b \in \tilde{H}^2(\mathbb{C}P^\infty)$ be generators. Then by the Kunneth isomorphism

$$H^k(S^1 \times \mathbb{C}P^\infty, S^1 \times \{x_0\}) \cong (H^0(S^1) \otimes \tilde{H}^k(\mathbb{C}P^\infty)) \oplus (H^1(S^1) \otimes \tilde{H}^{k-1}(\mathbb{C}P^\infty))$$

so we have 2 classes, $\alpha = a \otimes b$ and $\beta = 1 \otimes b$. $|\alpha| = 3$ and $|\beta| = 2$. Note $a^2 = 0$ and b^k generates $H^{2k}(\mathbb{C}P^\infty)$.

Thus

$$\alpha \smile \alpha = (-1)^{|b||a|} \alpha^2 \otimes b^2 = 0 \otimes b^2 = 0$$

$$\alpha \smile \beta = (-1)^{|b||1|} a \otimes b^2 = a \otimes b^2$$

$$\beta \smile \beta = (-1)^{|b||1|} 1 \otimes b^2 = 1 \otimes b^2$$

Thus

$$H^*((S^1 \times \mathbb{C}P^\infty)/(S^1 \times \{x_0\})) = \frac{\mathbb{Z}[\alpha, \beta]}{(\alpha^2)}$$

Can be extended rather trivially to all coefficients. Basically only part we need to mess with is $H^*(\mathbb{C}P^\infty)$ everything else follows with $\mathbb{R}$ -Kunneth Formula. $\square$